

Conditional Probability

Section 1.5

Example 1.15

Consider an experiment that consists of testing two integrated circuits that come from the same silicon wafer, and observing in each case whether a circuit is accepted (a) or rejected (r). The sample space of the experiment is $S = \{rr, ra, ar, aa\}$. Let B denote the event that the first chip tested is rejected. Mathematically, $B = \{rr, ra\}$. Similarly, let $A = \{rr, ar\}$ denote the event that the second circuit is a failure.

The circuits come from a high-quality production line. Therefore the prior probability $P[A]$ is very low. In advance, we are pretty certain that the second circuit will be accepted. However, some wafers become contaminated by dust, and these wafers have a high proportion of defective chips. Given the knowledge of event B that the first chip was rejected, our knowledge of the quality of the second chip changes. With the event B that the first chip is a reject, the probability $P[A|B]$ that the second chip will also be rejected is higher than the *a priori* probability $P[A]$ because of the likelihood that dust contaminated the entire wafer.

Definition 1.6 Conditional Probability

The conditional probability of the event A given the occurrence of the event B is

$$P[A|B] = \frac{P[AB]}{P[B]}.$$

Theorem 1.9

A conditional probability measure $P[A|B]$ has the following properties that correspond to the axioms of probability.

Axiom 1: $P[A|B] \geq 0$.

Axiom 2: $P[B|B] = 1$.

Axiom 3: If $A = A_1 \cup A_2 \cup \dots$ with $A_i \cap A_j = \emptyset$ for $i \neq j$, then

$$P[A|B] = P[A_1|B] + P[A_2|B] + \dots$$

Example 1.16 Problem

With respect to Example 1.15, consider the a priori probability model

$$P[rr] = 0.01, \quad P[ra] = 0.01, \quad P[ar] = 0.01, \quad P[aa] = 0.97. \quad (1.22)$$

Find the probability of $A =$ “second chip rejected” and $B =$ “first chip rejected.” Also find the conditional probability that the second chip is a reject given that the first chip is a reject.

Example 1.16 Solution

We saw in Example 1.15 that A is the union of two disjoint events (outcomes) rr and ar . Therefore, the a priori probability that the second chip is rejected is

$$P[A] = P[rr] + P[ar] = 0.02 \quad (1.23)$$

This is also the a priori probability that the first chip is rejected:

$$P[B] = P[rr] + P[ra] = 0.02. \quad (1.24)$$

The conditional probability of the second chip being rejected given that the first chip is rejected is, by definition, the ratio of $P[AB]$ to $P[B]$, where, in this example,

$$P[AB] = P[\text{both rejected}] = P[rr] = 0.01 \quad (1.25)$$

Thus

$$P[A|B] = \frac{P[AB]}{P[B]} = 0.01/0.02 = 0.5. \quad (1.26)$$

The information that the first chip is a reject drastically changes our state of knowledge about the second chip. We started with near certainty, $P[A] = 0.02$, that the second chip would not fail and ended with complete uncertainty about the quality of the second chip, $P[A|B] = 0.5$.

Theorem 1.10 Law of Total Probability

For an event space $\{B_1, B_2, \dots, B_m\}$ with $P[B_i] > 0$ for all i ,

$$P[A] = \sum_{i=1}^m P[A|B_i] P[B_i].$$

Example 1.19 Problem

A company has three machines B_1 , B_2 , and B_3 for making $1\text{ k}\Omega$ resistors. It has been observed that 80% of resistors produced by B_1 are within $50\text{ }\Omega$ of the nominal value. Machine B_2 produces 90% of resistors within $50\text{ }\Omega$ of the nominal value. The percentage for machine B_3 is 60%. Each hour, machine B_1 produces 3000 resistors, B_2 produces 4000 resistors, and B_3 produces 3000 resistors. All of the resistors are mixed together at random in one bin and packed for shipment. What is the probability that the company ships a resistor that is within $50\text{ }\Omega$ of the nominal value?

Example 1.19 Solution

Let $A = \{\text{resistor is within } 50 \, \Omega \text{ of the nominal value}\}$. Using the resistor accuracy information to formulate a probability model, we write

$$P[A|B_1] = 0.8, \quad P[A|B_2] = 0.9, \quad P[A|B_3] = 0.6 \quad (1.29)$$

The production figures state that $3000 + 4000 + 3000 = 10,000$ resistors per hour are produced. The fraction from machine B_1 is $P[B_1] = 3000/10,000 = 0.3$. Similarly, $P[B_2] = 0.4$ and $P[B_3] = 0.3$. Now it is a simple matter to apply the law of total probability to find the accuracy probability for all resistors shipped by the company:

$$P[A] = P[A|B_1] P[B_1] + P[A|B_2] P[B_2] + P[A|B_3] P[B_3] \quad (1.30)$$

$$= (0.8)(0.3) + (0.9)(0.4) + (0.6)(0.3) = 0.78. \quad (1.31)$$

For the whole factory, 78% of resistors are within $50 \, \Omega$ of the nominal value.

Theorem 1.11 Bayes' theorem

$$P[B|A] = \frac{P[A|B] P[B]}{P[A]}.$$

In Example 1.19, $P[B3] = 0.3$, $P[A]=0.78$, $P[A|B3] = 0.6$. Find the probability that an acceptable resistor comes from machine B3.

Example 1.20 Solution

Now we are given the event A that a resistor is within $50\ \Omega$ of the nominal value, and we need to find $P[B_3|A]$. Using Bayes' theorem, we have

$$P[B_3|A] = \frac{P[A|B_3] P[B_3]}{P[A]}. \quad (1.34)$$

Since all of the quantities we need are given in the problem description, our answer is

$$P[B_3|A] = (0.6)(0.3)/(0.78) = 0.23. \quad (1.35)$$

Similarly we obtain $P[B_1|A] = 0.31$ and $P[B_2|A] = 0.46$. Of all resistors within $50\ \Omega$ of the nominal value, only 23% come from machine B_3 (even though this machine produces 30% of all resistors). Machine B_1 produces 31% of the resistors that meet the $50\ \Omega$ criterion and machine B_2 produces 46% of them.

Independence

Section 1.6

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Independence

Definition 1.7 Two Independent Events

Events A and B are independent if and only if

$$P[AB] = P[A] P[B].$$

1.6 Comment: Independent vs. Disjoint

Keep in mind that **independent** and **disjoint** are *not* synonyms.

In some contexts these words can have similar meanings, but this is not the case in probability. Disjoint events have no outcomes in common and therefore $P[AB] = 0$. In most situations independent events are not disjoint! Exceptions occur only when $P[A] = 0$ or $P[B] = 0$. When we have to calculate probabilities, knowledge that events A and B are *disjoint* is very helpful. Axiom 3 enables us to *add* their probabilities to obtain the probability of the *union*. Knowledge that events C and D are *independent* is also very useful. Definition 1.7 enables us to *multiply* their probabilities to obtain the probability of the *intersection*.

Example 1.8

A short-circuit tester has a red light to indicate that there is a short circuit and a green light to indicate that there is no short circuit. Consider an experiment consisting of a sequence of three tests. In each test the observation is the color of the light that is on at the end of a test. An outcome of the experiment is a sequence of red (r) and green (g) lights. We can denote each outcome by a three-letter word such as rgr , the outcome that the first and third lights were red but the second light was green. We denote the event that light n was red or green by R_n or G_n . The event $R_2 = \{grg, grr, rrg, rrr\}$. We can also denote an outcome as an intersection of events R_i and G_j . For example, the event $R_1G_2R_3$ is the set containing the single outcome $\{rgr\}$.

Example 1.21 Problem

Suppose that for the three lights of Example 1.8, each outcome (a sequence of three lights, each either red or green) is equally likely. Are the events R_2 that the second light was red and G_2 that the second light was green independent? Are the events R_1 and R_2 independent?

Example 1.21 Solution

Each element of the sample space

$$S = \{rrr, rrg, rgr, rgg, grr, grg, ggr, ggg\} \quad (1.37)$$

has probability $1/8$. Each of the events

$$R_2 = \{rrr, rrg, grr, grg\} \quad \text{and} \quad G_2 = \{rgr, rgg, ggr, ggg\} \quad (1.38)$$

contains four outcomes so $P[R_2] = P[G_2] = 4/8$. However, $R_2 \cap G_2 = \emptyset$ and $P[R_2 G_2] = 0$. That is, R_2 and G_2 must be disjoint because the second light cannot be both red and green. Since $P[R_2 G_2] \neq P[R_2]P[G_2]$, R_2 and G_2 are not independent. Learning whether or not the event G_2 (second light green) occurs drastically affects our knowledge of whether or not the event R_2 occurs. Each of the events $R_1 = \{rgg, rgr, rrg, rrr\}$ and $R_2 = \{rrg, rrr, grg, grr\}$ has four outcomes so $P[R_1] = P[R_2] = 4/8$. In this case, the intersection $R_1 \cap R_2 = \{rrg, rrr\}$ has probability $P[R_1 R_2] = 2/8$. Since $P[R_1 R_2] = P[R_1]P[R_2]$, events R_1 and R_2 are independent. Learning whether or not the event R_2 (second light red) occurs does not affect our knowledge of whether or not the event R_1 (first light red) occurs.

Definition 1.8 *3 Independent Events*

A_1 , A_2 , and A_3 are independent if and only if

- (a) A_1 and A_2 are independent,
- (b) A_2 and A_3 are independent,
- (c) A_1 and A_3 are independent,
- (d) $P[A_1 \cap A_2 \cap A_3] = P[A_1]P[A_2]P[A_3]$.

Example 1.23 Problem

In an experiment with equiprobable outcomes, the event space is $S = \{1, 2, 3, 4\}$. $P[s] = 1/4$ for all $s \in S$. Are the events $A_1 = \{1, 3, 4\}$, $A_2 = \{2, 3, 4\}$, and $A_3 = \emptyset$ independent?

Example 1.23 Solution

These three sets satisfy the final condition of Definition 1.8 because $A_1 \cap A_2 \cap A_3 = \phi$, and

$$P[A_1 \cap A_2 \cap A_3] = P[A_1] P[A_2] P[A_3] = 0. \quad (1.44)$$

However, A_1 and A_2 are not independent because, with all outcomes equiprobable,

$$P[A_1 \cap A_2] = P[\{3, 4\}] = 1/2 \neq P[A_1] P[A_2] = 3/4 \times 3/4. \quad (1.45)$$

Hence the three events are dependent.

More than Two Independent

Definition 1.9 Events

If $n \geq 3$, the sets $A_1, A_2, \dots, A_n$ are independent if and only if

- (a) every set of $n - 1$ sets taken from $A_1, A_2, \dots, A_n$ is independent,*
- (b) $P[A_1 \cap A_2 \cap \dots \cap A_n] = P[A_1]P[A_2] \cdots P[A_n]$.*